

Multi-Armed Bandit

Action $\rightarrow$ Immediate reward

We want: choose action $\rightarrow$ maximize expected immediate reward

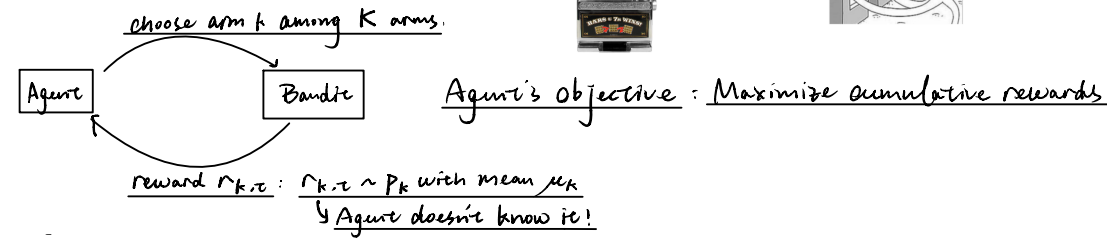
Learning to act in non-sequential setups

State $\rightarrow$ Agent: Act $\rightarrow$ Reward $\rightarrow$ End! No Next states!

if: state 1 $\rightarrow$ Act $\rightarrow$ Reward $\rightarrow$ State 2 $\Rightarrow$ not a stateless setup

Multi-Armed Bandit

At time t :



Action-value Function:

$Q_k(a) = \mathbb{E}[r_t | a_t = a]$ "The expected reward of choosing action a at time t "

$\Rightarrow$ If we know the arm with highest mean reward: just keep play that arm!
 (Actually we don't know)

Bernoulli Distribution

$$\text{Bern}(x|\mu) = \mu^x (1-\mu)^{1-x} \Rightarrow \begin{cases} x=1: p=\mu \\ x=0: p=1-\mu \end{cases}$$

$$\mathbb{E}[x] = \mu \quad \text{Var}[x] = \mu(1-\mu)$$

Gaussian Distribution:

$$\text{Norm}(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Basic Concept in MAB

1 Estimate action-value function

$Q_k(a)$ & $Q_k^*(a)$

$$Q_k = \frac{r_1 + r_2 + \dots + r_t}{t-1} \xrightarrow{\text{do it incrementally}} Q_{k,t+1} = Q_k + \frac{1}{t} [r_t - Q_k]$$

$$Q_{k,t+1} = \frac{r_1 + r_2 + \dots + r_t}{t} = \frac{1}{t} (r_t + \sum_{i=1}^{t-1} r_i) = \frac{1}{t} (r_t + (t-1) \frac{1}{t-1} \sum_{i=1}^{t-1} r_i) \\ = \frac{1}{t} (r_t + (t-1) Q_k) = Q_k + \frac{1}{t} [r_t - Q_k]$$

$\Rightarrow$ Do not need to store all r_i

Exploration/Exploitation Dilemma

$Q_k(a) \approx Q_k^*(a) \forall a$. estimate $Q_k(a)$.

$a_t^* = \arg \max_a Q_k(a)$ "Greedy action": a with highest $Q_k(a)$

if we choose "Greedy action a_t^* ": exploitation (make best decision given current info)
 if not: exploration (gather more info)

- The best long-term strategy may involve short-term sacrifices
- Gather enough information to make the best overall decisions
- The exploration/exploitation dilemma** is not a problem encountered in computational RL or deep RL: It is a fundamental problem in the decision-making of any intelligent agent.
- High-level idea:** You can never stop exploring, but maybe you should explore less over time

Action value: $Q_k(a) = \mathbb{E}[r_t | a_t = a] = \frac{r_1 + r_2 + \dots + r_t}{t}$

Optimal value $U_k = \max_a Q^*(a) = Q^*(a^*)$ "Greedy Action"

Regret: "Expected Opportunity loss for one step"

$$r_t = \mathbb{E}[U_k - Q_k(a_t)]$$

action given by alg at time t : random

Total Regret: $L_T = \sum_{t=1}^T l_t$ sum of regret at each time t

Maximize cumulative expected reward = minimize total regret.

Action Selection in MAB

Fixed Exploration Period + Greedy

- Allocate a fixed time period to exploration when you try bandits uniformly at random
- Estimate mean rewards for all actions:

$$Q_k(a) = \frac{1}{N_k(a)} \sum_{i=1}^{t-1} r_i \cdot \mathbb{I}(a_i = a)$$

- Select the action that is optimal for the estimated mean rewards, i.e., $a_t = \arg \max_a Q_k(a)$

$\Rightarrow$ will choose greedy action forever! $\Rightarrow$ linear total regret
 i.e. always explore

How to keep exploration?

ϵ -greedy prob: $\frac{\epsilon}{|\mathcal{A}|}$ $\arg \max_a Q_k(a)$ (greedy)

prob: ϵ
 $\rightarrow$ random action (exploration)

A simple bandit algorithm

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Initialize, for  $a = 1$  to  $k$ :
   $Q(a) \leftarrow 0$ 
   $N(a) \leftarrow 0$ 

Repeat forever:
   $A \leftarrow \begin{cases} \arg \max_a Q(a) & \text{with probability } 1 - \epsilon \\ \text{a random action} & \text{with probability } \epsilon \end{cases}$  (breaking ties randomly)
   $R \leftarrow \text{bandit}(A)$ 
   $N(A) \leftarrow N(A) + 1$ 
   $Q(A) \leftarrow Q(A) + \frac{1}{N(A)} [R - Q(A)]$ 
    
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* ϵ -greedy still has linear total regret!

$$L_t \geq \frac{\epsilon}{|\mathcal{A}|} \sum_{a \neq a^*} \Delta_a \quad \text{where } \Delta_a = \mu^* - \mu_a, \mu_a \text{ is the second best outcome}$$

Decaying ϵ -Greedy

$$a_1, a_2, \dots, a_n \\ \xrightarrow{\text{decaying}}$$

Upper Confidence Bounds

Uncertainty: we don't know the exact expected reward of a bandit

The more we're uncertain $\rightarrow$ The more we should explore it! : May turn out to better action

UCB

$$Q_k(a) \leq \underbrace{\hat{Q}_k(a)}_{\text{estimated mean}} + \underbrace{U_k(a)}_{\text{"UCB"}} \rightarrow \text{"UCB"}$$

We should select: $a_t = \arg \max_a Q_k(a) + U_k(a)$

"UC" should satisfy: $N_k(a) \uparrow \Rightarrow U_k(a) \downarrow$ (uncertain)
 $N_k(a) \uparrow \Rightarrow U_k(a) \downarrow$ (accurate)
 "# tries of a "

Hoeffding's Inequality

$Z_1, \dots, Z_n$ be independent bounded RV $Z_i \in [a, b]$

$$\begin{cases} p\left(\frac{1}{n} \sum_{i=1}^n Z_i - \mathbb{E}[Z_i] \geq \tau\right) \leq \exp\left(-\frac{2n\tau^2}{(b-a)^2}\right) \\ p\left(\frac{1}{n} \sum_{i=1}^n Z_i - \mathbb{E}[Z_i] \leq -\tau\right) \leq \exp\left(-\frac{2n\tau^2}{(b-a)^2}\right) \end{cases} \quad \text{for all } \tau \geq 0$$

① $Q_k \in [0, 1] \Rightarrow b-a=1$.

$$\begin{cases} p(Q_k(a) - Q^*(a) \geq U_k(a)) \leq \exp(-2N_k(a)U_k(a)^2) \\ p(Q_k(a) - Q^*(a) \leq -U_k(a)) \leq \exp(-2N_k(a)U_k(a)^2) \end{cases}$$

$\downarrow$ transform.

$$\begin{cases} p(Q_k(a) - U_k(a) \geq Q^*(a)) \leq \exp(-2N_k(a)U_k(a)^2) \\ p(Q_k(a) + U_k(a) \leq Q^*(a)) \leq \exp(-2N_k(a)U_k(a)^2) \end{cases}$$

- We will apply Hoeffding's Inequality to the rewards obtained from each action (bandit) a :

$$p(Q_k(a) - U_k(a) \geq Q^*(a)) \leq c \cdot \exp(-2N_k(a)U_k(a)^2)$$

$$p(Q_k(a) + U_k(a) \leq Q^*(a)) \leq c \cdot \exp(-2N_k(a)U_k(a)^2)$$

- t : how many times I have played any action
- $N_t(a)$: how many times I have played the action a in t interactions
- $U_t(a)$: made the margin to depend on the amount of interactions t

$$\Rightarrow e^{-2N_k(a)U_k(a)^2} = p$$

$$U_k(a) = \sqrt{\frac{-\log p}{2N_k(a)}}$$

We want: $p \rightarrow 0$ as $t \rightarrow \infty$

We can let: $p = t^{-k}$

$$U_k(a) = \sqrt{\frac{-\log t^{-k}}{2N_k(a)}} = \sqrt{\frac{k \log t}{2N_k(a)}}$$

(let $k \geq 2$) $\Rightarrow U_k(a) = \sqrt{\frac{\log t}{N_k(a)}}$
 k can be 1, 2, 3, 4, ...

$$\Rightarrow a_t = \arg \max_a [Q_k(a) + U_k(a)]$$

$$= \arg \max_a \left[Q_k(a) + c \sqrt{\frac{\log t}{N_k(a)}} \right]$$

"c" is a "trade-off" between exploration & exploitation

- CB grows with the total number of actions we have taken t but shrinks with the number of times we have tried this particular action $N_t(a)$.
- This ensures each action is tried infinitely often but still balances exploration/exploitation.

Gradient Bandit

How about a policy-based solution?

m score parameters one for each action: $\{s_a | a \in \mathcal{A}\}$

$\pi: \mathcal{A} \rightarrow [0, 1]$: action selection probabilities

$$\pi(a) = \frac{e^{s_a}}{\sum_{b \in \mathcal{A}} e^{s_b}} \quad (\text{softmax})$$

$$J(s_a, s_{a'}, \dots, s_{a_n}) = \sum_{a \in \mathcal{A}} \pi(a) \mathbb{E}[r | a]$$

"Expected Reward"

Calculate Gradient of $J()$

$$\frac{\partial J}{\partial s_a} = \sum_{a' \in \mathcal{A}} \mathbb{E}[r | a'] \frac{\partial \pi(a')}{\partial s_a}$$

$$\frac{\partial \pi(a')}{\partial s_a} = \pi(a') \cdot \frac{1}{\pi(a')} \cdot \frac{\partial \pi(a')}{\partial s_a}$$

$$= \pi(a') \cdot \frac{\partial (\log \pi(a'))}{\partial \pi(a')} \cdot \frac{\partial \pi(a')}{\partial s_a}$$

$$= \pi(a') \cdot \frac{\partial (\log \pi(a'))}{\partial s_a}$$

$$\Rightarrow \frac{\partial J}{\partial s_a} = \sum_{a' \in \mathcal{A}} \mathbb{E}[r | a'] \pi(a') \cdot \frac{\partial (\log \pi(a'))}{\partial s_a}$$

$$= \mathbb{E}_{a' \sim \pi} [r \cdot \frac{\partial (\log \pi(a'))}{\partial s_a}]$$

$$\begin{aligned} &= \left[\frac{\partial \log \frac{e^{s_{a'}}}{\sum_{b \in \mathcal{A}} e^{s_b}}}{\partial s_a} \right] = \frac{\partial [s_{a'} - \log(e^{s_1} + e^{s_2} + \dots + e^{s_n})]}{\partial s_a} \\ &= \frac{\partial s_{a'}}{\partial s_a} - \frac{\partial \log(e^{s_1} + e^{s_2} + \dots + e^{s_n})}{\partial s_a} \\ &= 1(a' = a) - \frac{1}{(e^{s_1} + e^{s_2} + \dots + e^{s_n})} \cdot e^{s_a} \\ &= 1(a' = a) - \pi(a) \end{aligned}$$

$$\frac{\partial J}{\partial s_a} = \mathbb{E}_{a' \sim \pi} [r \cdot (1(a' = a) - \pi(a))]$$